SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Case Study

Course Code:	ITA0517	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)*		
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Section / Batch:	BTECH IT	Date:	12\08\2026

Code of Conduct

*I **RAJA RAJESHWARI M (192421253)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **RAJA RAJESHWARI M**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study
COURSE NAME: COMPUTER VISION COURSE CODE:ITA0517
NAME:RAJA RAJESHWARI M REG NO:192421253
COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)
Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision
techniques. Justify your answers with relevant reasoning and comparisons wherever
necessary.

7. Corner Detection vs Edge Detection in Tracking
1. Understanding of the Case
A tracking system uses edge detection to identify the boundaries of moving objects. However, edges alone are insufficient for accurate tracking across consecutive video frames.

- **Edge detection identifies areas of sudden intensity change, such as object boundaries.**
- **An edge generally contains information mainly about one direction (the direction perpendicular to the edge).**
- **Long, straight edges may look very similar at different locations, making it difficult to determine the exact movement of an object.**
- **Edges can also be affected by lighting changes, shadows, noise, occlusion, and motion blur.**
- **Therefore, the tracker may lose the object's position or incorrectly match an edge from one frame to another.**

2. Why Edges Are Insufficient
Consider a moving rectangular object:
Frame 1   Frame 2

```
+---------+              +---------+
|         |     →        |         |
| Object  |              | Object  |
|         |              |         |
+---------+              +---------+
```

Edge detection mainly detects the four boundaries:

```
┌────────────────┐
│      │
│      │
└────────────────┘
```

The problem is that the boundaries are not highly distinctive. A tracker may find many similar edge points in the next frame and cannot reliably determine which point corresponds to the previous one.
Thus:

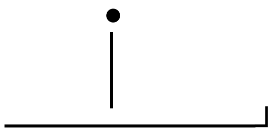
Edges → good for detecting boundaries, but weak for establishing unique point-to-point correspondence.

3. Why Corner Detection Is Better for Tracking

A corner occurs where two or more edges meet and the image intensity changes significantly in more than one direction.

For example:

Corner



Corners contain much more distinctive local information than straight edges.

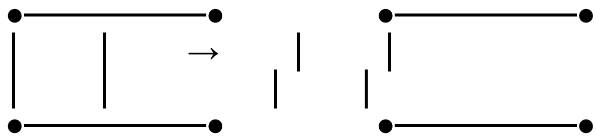
Common corner-detection methods include:

- Harris Corner Detector
- Shi-Tomasi Corner Detector
- FAST (Features from Accelerated Segment Test)

A corner can be represented as a feature point and tracked from one frame to another.

For example:

Frame 1 Frame 2



The four corner points provide identifiable locations that can be matched between frames.

4. Technical Comparison

Feature	Edge Detection	Corner Detection
Detects	Object boundaries	Distinctive feature points
Information	Mainly one direction	Multiple directions
Uniqueness	Low	High
Tracking suitability	Limited	Excellent
Point correspondence	Difficult	Easier
Sensitivity to motion	Can cause ambiguity	More reliable for feature tracking
Example algorithms	Canny, Sobel	Harris, Shi-Tomasi, FAST
Main use	Segmentation/boundary detection	Tracking and feature matching

5. Analysis

The main issue with using only edges is the aperture problem.

For a straight edge, movement along the edge cannot be determined accurately because many neighboring points have almost identical appearance.

For example:
Edge:

Possible movement:
← ● →
The tracker cannot determine whether the point moved left or right along the edge.
At a corner, however, intensity changes occur in two directions:

——————●

Therefore, the movement of the corner can be estimated much more accurately.
Mathematically, a good tracking point should have significant intensity variation in both image directions:
where and are the eigenvalues of the local structure/gradient matrix.
For a strong corner:
This means the feature contains sufficient information to determine its movement in both the x and y directions.

6. Proposed Solution
A more reliable tracking system should combine corner detection with optical-flow or feature-matching techniques.
Proposed Tracking Pipeline
Input Video
↓
Pre-processing
↓
Corner Detection
(Harris / Shi-Tomasi)
↓
Select Strong Feature Points
↓
Track Features Across Frames
(Lucas-Kanade Optical Flow)
↓
Remove Incorrect / Lost Features
↓
Estimate Object Motion
↓
Reliable Object Tracking
Improvements

1. Apply Gaussian filtering before corner detection to reduce noise.
2. Use Shi-Tomasi or Harris corner detection to identify strong feature points.
3. Track these points using Lucas-Kanade optical flow.
4. Reject incorrect matches using a distance/error threshold.
5. Use RANSAC when necessary to remove outlier correspondences.
6. Detect new corners when existing tracking points disappear.

7. Combine feature tracking with an object detector when the scene contains severe occlusion or objects entering/leaving the frame.

7. Expected Result

Using corners instead of only edges provides:

- **More distinctive tracking points.**
- **Better frame-to-frame correspondence.**
- **Improved motion estimation.**
- **Greater robustness to object movement.**
- **Reduced ambiguity along straight boundaries.**
- **More reliable tracking even when objects undergo moderate rotation or translation.**

Conclusion

Edge detection is useful for identifying object boundaries, but it does not provide sufficiently distinctive points for reliable tracking. Corners are better tracking features because they contain intensity changes in multiple directions and can therefore be localized and matched more accurately between consecutive frames.